

# Designing a Heat Recovery System for Cooper

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## Abstract

This project compares a double-pipe heat exchanger and a plate heat exchanger for recovering heat from exhaust air in Cooper Union's 41 Cooper Square HVAC system. The system was modeled for winter operation, with warm indoor exhaust air entering at 20°C and cold outdoor supply air entering at 0°C. The cold-side air flow rate was taken as 4.5 kg/s, and the heat exchangers were compared using energy balances, convection correlations, the NTU-effectiveness method, and the log mean temperature difference method. A fixed heat transfer area of 10 m<sup>2</sup> was used for the main design comparison. For counterflow operation at a hot-side mass flow rate of 1 kg/s, the plate heat exchanger transferred about 18,000 W of heat with an effectiveness of 0.74, raising the cold outlet temperature to about 4.1°C. The selected plate design used 34 plates and fit within an approximate 1.5 m × 1 m × 1 m envelope. The double-pipe heat exchanger transferred about 20,000 W with an effectiveness of 0.81, but required about 32 m of pipe, arranged as four 4 m hairpins, with an approximate envelope of 5 m × 1 m × 1 m. The plate heat exchanger provided similar performance compared to the double pipe heat exchanger in a much smaller and more scalable size. Because Cooper Union is in New York City, where mechanical room space is limited, the more compact plate heat exchanger was the better option for this application. This analysis is limited by the assumptions of constant air properties, no fouling, negligible heat loss to the surroundings, and simplified pressure-drop and fan power effects.

## Nomenclature

| Parameter       | Units              | Meaning                                                                  |
|-----------------|--------------------|--------------------------------------------------------------------------|
| $q$             | $W$                | Heat transfer rate                                                       |
| $\dot{m}$       | $\frac{kg}{s}$     | Mass flow rate of the chosen fluid                                       |
| $c_p$           | $\frac{J}{K * kg}$ | Specific heat capacity                                                   |
| $C$             | $\frac{J}{K * s}$  | Heat capacity rate                                                       |
| $R = C_r$       | $\sim$             | Ratio of minimum to maximum heat capacity rate                           |
| $U$             | $\frac{W}{m^2 K}$  | Overall heat transfer coefficient                                        |
| $A$             | $m^2$              | Heat transfer area                                                       |
| $k$             | $\frac{W}{m * K}$  | Thermal conductivity                                                     |
| $\mu$           | $\frac{kg}{m * s}$ | Dynamic Viscosity                                                        |
| $NTU$           | $\sim$             | Representation of the heat exchanger size relative to the fluid capacity |
| $\epsilon$      | $\sim$             | Effectiveness of the heat exchanger                                      |
| $T_{h,i}$       | $K$                | Inlet temperature of hot gas                                             |
| $T_{c,i}$       | $K$                | Inlet temperature of cold gas                                            |
| $h$             | $\frac{W}{m^2 K}$  | Convection heat transfer coefficient                                     |
| $\Delta T_{lm}$ | $K$                | Log Mean Temperature Difference (LMTD)                                   |
| $D_1$           | $m$                | Inner diameter of outer pipe                                             |
| $D_i$           | $m$                | Inner diameter of inner pipe                                             |
| $D_o$           | $m$                | Outer diameter of inner pipe                                             |
| $h_h$           | $\frac{W}{m^2 K}$  | Convection heat transfer coefficient of hot fluid                        |
| $h_c$           | $\frac{W}{m^2 K}$  | Convection heat transfer coefficient of cold fluid                       |
| $N_p$           | $\sim$             | Number of Plates                                                         |
| $A_p$           | $m^2$              | Area of plate                                                            |
| $W_p$           | $m$                | Width of plate                                                           |
| $C_h$           | $\sim$             | Constant for Nusselt Number Calculation                                  |
| $L_{hairpin}$   | $m$                | Length or hairpin used                                                   |
| $t$             | $m$                | Thickness of material used                                               |
| $b$             | $m$                | Plate gap distance                                                       |

# 1. Introduction

## 1.1 Objective and Scope

Heating, ventilation, and air-conditioning (HVAC) systems account for a significant portion of energy consumption in modern buildings. During ventilation, conditioned indoor air is exhausted from the building and replaced with outdoor air, which can represent a major source of energy loss during heating seasons. Heat recovery systems are therefore used to capture thermal energy from the exhaust air stream and transfer it to the incoming outdoor air, reducing the overall heating demand and improving energy efficiency. Cooper Union's foundation building, despite its modern HVAC and Building Management System, still faces pressure to reduce building emissions under New York City's Carbon Challenge and Local Law 97. The Carbon Challenge is a voluntary city program that encourages institutions to reduce building emissions by campus, while Local Law 97 sets mandatory carbon limits for large buildings and can result in penalties if those limits are exceeded. A major portion of Cooper Union's emissions come from Air Handling Units (AHUs), which are part of the HVAC system and are responsible for filtering, conditioning, and circulating air to maintain indoor air quality and comfort [1].

Introducing a heat recovery system through means of a Heat Exchanger would help to decrease the amount of energy exhausted heating the outdoor air that AHU 6 takes to control air in floors 3 and above Cooper Square. A double pipe heat exchanger was initially chosen as the baseline choice for this analysis [2]. This design was chosen due to its easy maintenance and cleaning, it's cost effectiveness, and the reasonable heat transfer efficiency, which pairs well with Cooper's budget and needs. Therefore, the efficiency, heat transfer rate, cold outlet temperature, logarithmic mean temperature difference (LMTD) will be measured as a

function of the warm air mass flow rate and compared to a multiple plate heat exchanger, to determine the most suitable heat exchanger in terms of Heat Transfer quantities and feasibility [1].

## 2. Materials and Methods

### 2.1 Schematic

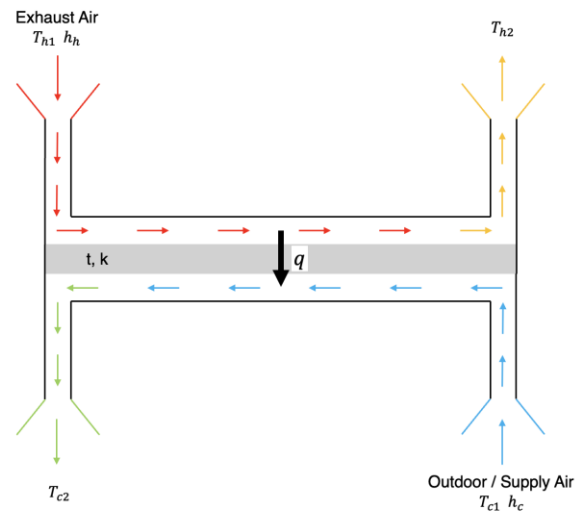


Figure 1: Diagram of Plate Heat Exchanger

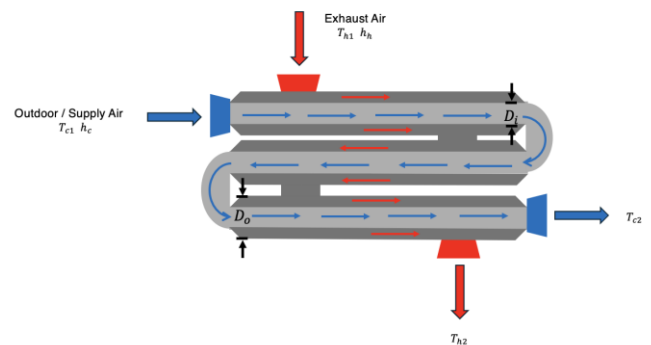


Figure 2: Diagram of Double Pipe Heat Exchanger

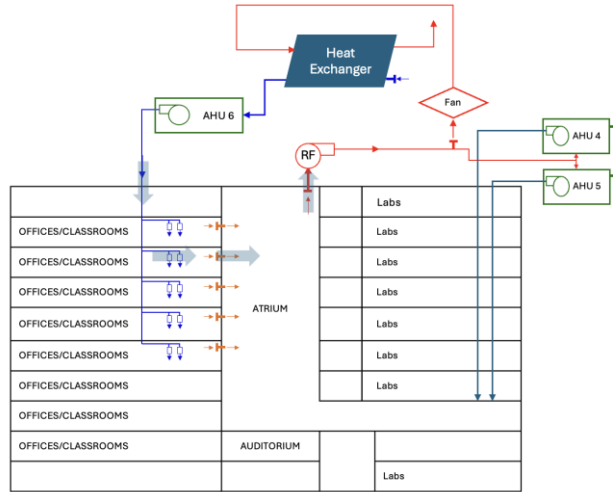


Figure 3: Diagram of Multiple Plate Heat Exchanger

## 2.2 Given Information

The inlet temperatures were determined, assuming that the warm inside air was at room temperature while the outside air was assumed to be 0°C for a winter day. Knowing the rate at which AHU extracts cold air also helped to determine the cold mass flow rate. For both the setups, the following givens were chosen.

Table 1: Chosen parameters applicable to both setups

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| $\dot{m}_{c,h}$ | $8000 \text{ CFM} \sim 4.5 \frac{\text{kg}}{\text{s}}$ |
| $c_p$           | $1,006 \frac{\text{J}}{\text{kg} * \text{K}}$          |
| $T_{h,i}$       | $293.15 \text{ K}$                                     |
| $T_{c,i}$       | $273.15 \text{ K}$                                     |
| $\mu$           | $1.81 * 10^5 \text{ Pa} * \text{s}$                    |

Table 2: Chosen inputs applicable to Plate Heat Exchanger

|       |                   |
|-------|-------------------|
| $A_p$ | $0.3 \text{ m}^2$ |
| $W_p$ | $0.3 \text{ m}$   |
| $b$   | $0.01 \text{ m}$  |
| $C_h$ | $0.4$             |
| $A$   | $10 \text{ m}^2$  |
| $F$   | $1.15$            |

|       |                                           |
|-------|-------------------------------------------|
| $k_p$ | $15 \frac{\text{W}}{\text{m} * \text{K}}$ |
| $t_p$ | $0.006 \text{ m}$                         |
| $L_p$ | $1 \text{ m}$                             |

Table 3: Chosen parameters applicable to Double Heat Exchanger

|       |                       |
|-------|-----------------------|
| $L$   | <i>Length</i>         |
| $D_i$ | <i>Inner diameter</i> |
| $t$   | <i>Thickness</i>      |

## 2.3 Modeling Assumptions

For further simplicity in analysis the following assumptions were made.

1. Negligible heat loss to the surroundings.
2. Negligible kinetic and potential energy changes.
3. Constant properties.
4. Negligible fouling factors.
5. Fully developed conditions for the air

## 2.4 Intended Approach

The thermal analysis begins with a Hot & Cold Side Energy Balance to ensure conservation of energy across each side of the combined system. The heat transfer coefficients are determined through a scaling sequence of  $\text{Re} \rightarrow \text{Nu} \rightarrow \text{h} \rightarrow \text{U}$ . To optimize the design, an iterative LMTD and NTU approach was used to solve for the theoretical output over the parameterized geometry. Outlet temperatures were temporarily assumed, then updated by sweeping the data at each input, tracking the difference between heat transfers determined by each method until results converged. This was then followed by an efficiency comparison between double-pipe and plate heat exchangers to determine the most effective configuration. Parallel and counterflow arrangements were considered.

### 3. Results and Discussion

#### 3.1 General Heat Transfer Analysis (Akpan Ukemeobong E., 2025)

##### Energy Balance

According to Newton's Law of Energy Conservation

$$\dot{m} \left( h + \frac{v^2}{2} + gz \right)_{in} - \dot{m} \left( h + \frac{v^2}{2} + gz \right)_{out} + q - \dot{W} = 0$$

There is no shaft work, pump work, or change in potential energy of within the defined control volumes; the governing conservation of energy equation can be reduced to

$$\dot{m}(h)_{in} - \dot{m}(h)_{out} = q$$

Assuming that the air is an ideal gas

$$\Delta h = c_p \Delta T$$

Therefore, the heat transfer of the hot and cold sides is defined by the energy balance as

$$q_h = \dot{m}_h c_p (T_{h,i} - T_{h,o})$$

$$q_c = \dot{m}_c c_p (T_{c,o} - T_{c,i})$$

##### Outlet Temperature

The outlet temperatures at each side of the exchanger can be related using

$$T_{h,o} = T_{h,i} - \frac{q}{\dot{m}_h c_p},$$

$$T_{c,o} = T_{c,i} + \frac{q}{\dot{m}_c c_p}.$$

##### Heat Capacity Rate Model

The Heat Capacity Rate (C) is a measure of the amount of heat a given fluid can absorb or release per unit of temperature change and per unit of time, which makes this a major choice in Heat Exchanger design. and of the chosen fluid in units  $c_p$ , the quantity can be defined by

$$C = \dot{m} * c_p.$$

Of the two fluids undergoing heat exchange, one can have a lower heat capacity rate, or both can have the same heat capacity rate. This can be shown through

$$C_{min} = \min(C_h, C_c),$$

$$C_{max} = \max(C_h, C_c).$$

From here, the heat capacity ratio can be defined as

$$R \equiv C_r = \frac{C_{min}}{C_{max}}.$$

##### Maximum Heat Transfer (q)

The maximum heat transfer rate is defined by

$$q_{max} = C_{min}(T_{h,i} - T_{c,i})$$

##### NTU Effectiveness ( $\epsilon$ ) Analysis Model

NTU is a dimensionless parameter that measures the heat transfer capacity of a heat exchanger relative to the minimum heat capacity rate,  $C_{min}$ . This relationship is written as

$$NTU = \frac{U * A}{C_{min}}.$$

The effective heat transfer effectiveness is defined as

$$\epsilon = \frac{q}{q_{max}} = \frac{q}{C_{min}(T_{h,i} - T_{c,i})}$$

Effective heat transfer is related to maximum transfer by efficiency, where

$$q_{NTU} = \epsilon q_{max}$$

##### Log Mean Temperature Difference (LMTD)

The log mean temperature difference ( $\Delta T_{lm}$ ) represents the driving force for heat transfer, with larger  $\Delta T_{lm}$  signifying higher heat transfer and for the given heat transfer coefficient and surface area. The equation for this quantity is

$$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\left(\frac{\Delta T_2}{\Delta T_1}\right)}$$

The temperature differences,  $\Delta T_1$  and  $\Delta T_2$ , are

$$\Delta T_1 = T_{h,i} - T_{c,o},$$

$$\Delta T_2 = T_{h,o} - T_{c,i},$$

for counterflow, and

$$\Delta T_1 = T_{h,i} - T_{c,i},$$

$$\Delta T_2 = T_{h,o} - T_{c,o},$$

for parallel flow.

Heat transfer is defined as

$$q_{LMTD} = UA\Delta T_{lm}$$

LMTD and NTU Convergence

NTU and LMTD methods are related through heat transfer rate,

$$q_{NTU} = q_{LMTD}$$

Convergence between heat transfer models is used to solve the outlet temperatures for each given set of geometric parameters.

### 3.2 Double Pipe Heat Exchanger (CheGuide, 2015)

The cross-sectional area of the annular flow, or the region between the outer surface of the inner pipe and the inner surface of the outer pipe is

$$A_{annulus} = \frac{\pi}{4}(D_1^2 - D_o^2)$$

Heat transfer surface area is defined as

$$A = \pi D_o L$$

Specifically, for a double pipe heat exchanger in counterflow the effectiveness is

$$\varepsilon = \frac{1 - \exp[-NTU(1 - C_r)]}{1 - C_r \cdot \exp[-NTU(1 - C_r)]}, \quad C_r \neq 1$$

and

$$\varepsilon = \frac{NTU}{1 + NTU}, \quad C_r = 1.$$

For a parallel flow arrangement, the effectiveness can be calculated as

$$\varepsilon = \frac{1 - \exp[-NTU(1 + C_r)]}{1 + C_r}.$$

Assuming that there is no fouling, as this is a new heat exchanger, the overall heat transfer coefficient ( $U$ ) can then be written as

$$\frac{1}{U} = \frac{D_o}{D_i} \frac{1}{h_i} + \frac{D_o \ln\left(\frac{D_o}{D_i}\right)}{2k_t} + \frac{1}{h_o}$$

Convection coefficient  $h$  values are found using the Reynolds, and Nusselt Number Correlations. Here,

$$Re = \frac{4\dot{m}}{\pi D \mu} \quad Nu = \frac{hD}{k},$$

and

$$Nu = 0.023 * Re^{0.8} * Pr^n \text{ (Turbulent),}$$

$$Nu = 3.66 \text{ (Laminar),}$$

with the inner tube using  $D = D_i$  and the annular side using  $D = D_1 - D_o$ , and  $n$  being equal to 0.3 and 0.4 for a fluid being cooled and heated, respectively.  $Pr$  and  $\mu$  are also defined at the respective temperatures. Therefore

$$h = \frac{kNu}{D}$$

And the number of “hairpins” could be found using

$$N_{Hairpin} = \frac{L}{2 * L_{Hairpin}}$$

### 3.3 Plate Heat Exchanger (Chevron) (Fábio A.S. Mota, 2015)

For a plate heat exchanger arrangement with multiple stacked plates, the effectiveness for parallel flow is

$$\epsilon = \frac{1 - e^{-NTU(1 + C_r)}}{1 + C_r},$$

and for counterflow is

$$\epsilon = \frac{1 - e^{-NTU(1 - C_r)}}{1 - C_r e^{-NTU(1 - C_r)}}.$$

Again, assuming no fouling, the overall heat transfer coefficient can then be defined as

$$U = \frac{1}{\frac{1}{h_{hot}} + \frac{t_p}{k_p} + \frac{1}{h_{cold}}}.$$

These  $h$  values can be found using a similar process as with the double pipe arrangement with the hydraulic diameter for the and Reynold number being defined by

$$D_e = \frac{2b}{\phi}$$

Where  $\phi \approx 1$  and

$$Re = \frac{G_c D_e}{\mu} \quad \text{where} \quad G_c = \frac{\dot{m}}{b \cdot W_p}.$$

The Nusselt number correlations could then be defined as

$$Nu = \frac{h D_e}{k} = C_h (Re)^n Pr^{\frac{1}{3}} \left( \frac{\mu}{\mu_w} \right)^{0.17},$$

Using the  $n$  and  $C_h$  values corresponding to the  $\beta$ , as shown in the Appendix. The necessary surface area and number of plates require can also be calculated using

$$N_p = \frac{A}{A_p}.$$

### 3.4 Design Comparison

#### Heat Transfer Comparison

According to the plots in the Appendix which display the comparisons of heat rate, effectiveness, and cold outlet temperatures, for both counterflow and parallel flow, counterflow is the superior option. The temperature difference between the hot and cold fluids is more uniform across the entire heat transfer area. For parallel flow, the maximum temperature difference occurs at the start and exponentially decreases as the fluids move toward equilibrium.

To compare the two exchangers a set heat transfer area of  $10m^2$  will be used and they will be plotted over the unknown  $\dot{m}_h$ . The Plate Heat Exchanger uses AISI 316 Stainless Steel for the

Plate Heat Exchanger to ensure longevity against potential corrosive or damaging elements in the air. Using plates with  $L_p = 1 \text{ m}$ ,  $W_p = 0.3 \text{ m}$ , and  $t_p = 0.006 \text{ m}$  with  $b = 0.01 \text{ m}$  spacing, at  $\dot{m}_h = 4 \frac{kg}{s}$  for counterflow yields 18,000 W of heat transferred, at an effectiveness of 0.74. This means that a total of 34 plates can be used with a 1.5 m x 1 m x 1 m envelope, bring the outlet cold temperature to  $T_{c,o} = 4.1 \text{ }^\circ\text{C}$ .

The Double Pipe Heat Exchanger also uses AISI 316 Stainless Steel for the inner and outer pipes. The selected geometric parameters for the design comparison were based on an inner tube with a  $D_i = 0.1 \text{ m}$  diameter and 0.004 m thickness along with an outer tube diameter of  $D_1 = 0.15 \text{ m}$ . At  $\dot{m}_h = 4 \frac{kg}{s}$  for counterflow, about 20,000 W were transferred from the indoor air to outdoor air, at an effectiveness of 0.81. This also made  $T_{c,o} = 4.5 \text{ }^\circ\text{C}$ . The length of pipe to achieve this, however, was 32 m. While this can be condensed using  $L_{Hairpin} = 4 \text{ m}$  long hairpins, giving  $N_{Hairpins} = 4$ , the envelope of the heat exchanger would be around 5 m x 1 m x 1 m in size. The plots for these comparisons are shown in the Appendix.

Though double pipe heat exchangers are more easily maintained, and naturally made for counterflow, to achieve high heat transfer over a large heat transfer area makes the design bulky and not feasible to use in 41 Cooper Square. The pipe ends up having to be so long that multiple hairpins are needed. Also, with such a high mass flow rate for the cold air, such small pipes will experience quite high velocity. Plate Heat Exchangers, while occasionally more complex and sensitive in gasket regions, offer similar heat exchange in a much more compact space, which aligns with Cooper's needs. They are also very easily scalable, as the capacity of a gasketed plate unit could be increased by simply adding more plates.

A potential design change that could increase the heat transfer for both setups would be to increase the Heat Transfer Area. To accomplish this and increase effectiveness without making the heat exchanger larger, fins can be added to both designs. For the double pipe setup fins could be added to the outer walls of the inner pipe and the inner walls of the outer pipe. Adding corrugated fins between flat separator plates for the Plate Heat Exchanger could also accomplish this. Increasing the heat transfer area increases the NTU's of the heat exchanger and correspondingly increases its effectiveness. Air, unlike water, is a poor thermal conductor, and therefore increasing this parameter maximizes heat transfer and can help minimize exchanger envelope.

## 4. Conclusion

Using the selected geometry, the double pipe heat exchanger outperformed the plate heat exchanger with a higher heat transfer rate ( $20000W > 18000W$ ), higher effectiveness ( $0.81 > 0.74$ ) and higher cold outlet temperature ( $4.5^{\circ}C > 4.1^{\circ}C$ ). However, due to the large system envelope, this design may be impractical for a building like 41 Cooper Square. The limitation to this analysis is that pressure drop and power exhausted by the hot side fan were not accounted for in the baseline analysis. When considering pressure drop, the use of hairpins and specified plate spacings may require more exchangers to be needed to save more energy than the fan exhausts. This can be shown in the "Next Steps" portion of the Appendix. This demonstrates the engineering relevance and engineering challenges that come along with heat exchanger and recover design. While certain configurations provide better overall results, they may not be feasible for a given building. At the same time engineering simplifications such as ignoring pressure drops and fan power could be detrimental to energy efficiency.

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## Appendix

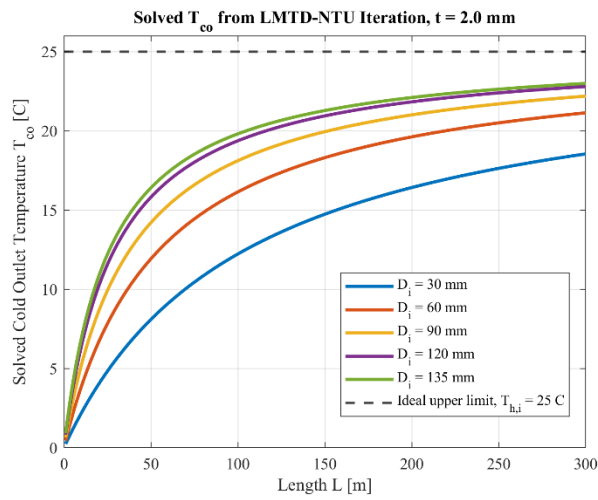


Figure 3: Solved  $T_{c,o}$  from LMTD-NTU Iteration

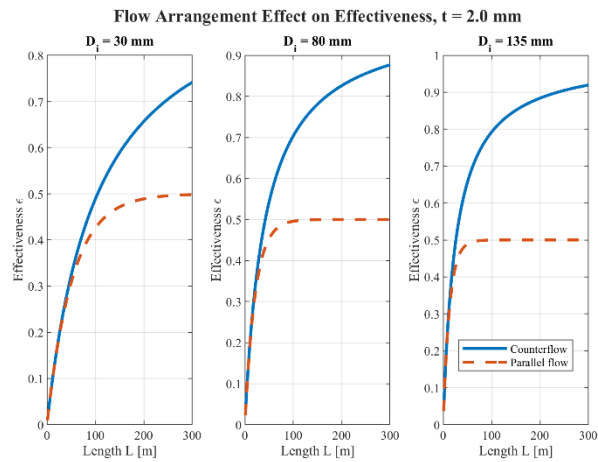


Figure 4: Flow Arrangement Effect on Effectiveness

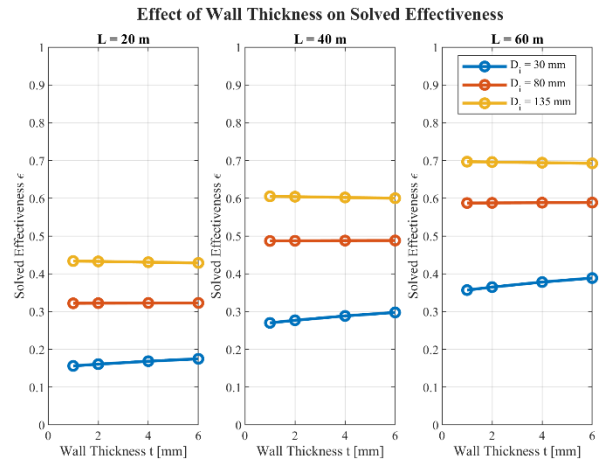


Figure 5: Effect of Wall Thickness on Solved Effectiveness

Counterflow:

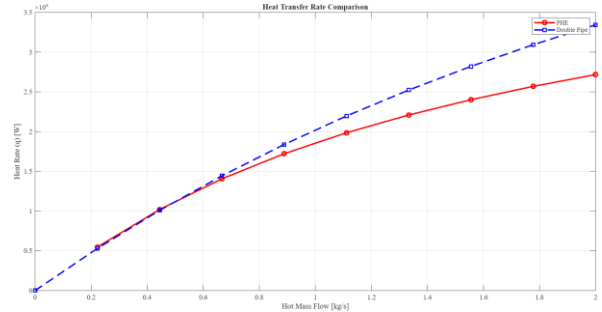


Figure 6: Heat Transfer Rate Comparison - Counterflow

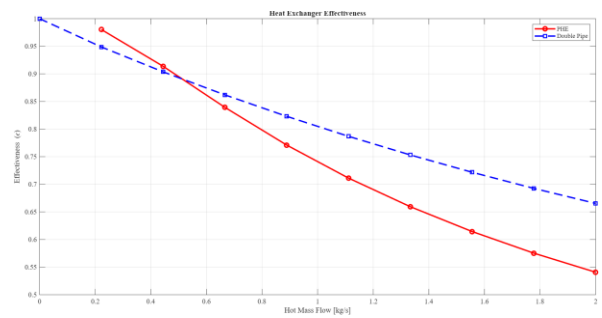


Figure 7: Heat Exchanger Effectiveness - Counterflow

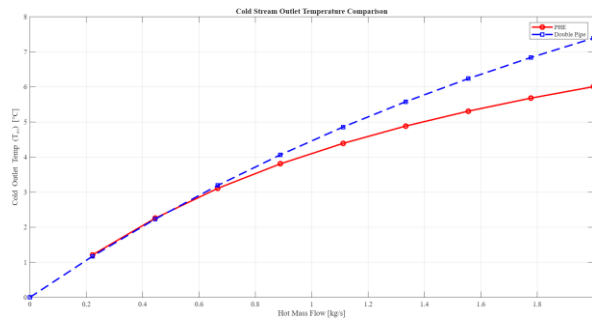


Figure 8: Cold Stream Outlet Temperature Comparison - Counterflow

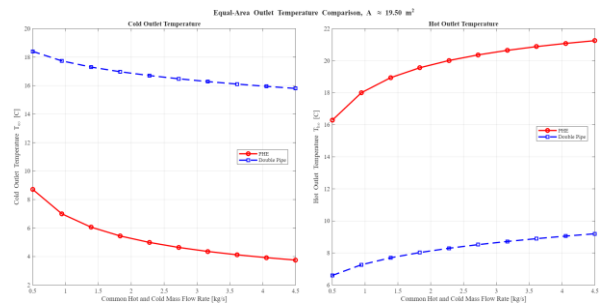


Figure 12: Equal-Area PHE, Double Pipe Outlet Temperature Comparison

## Next Steps and Future Considerations (Same Hot and Cold Flow Rate):

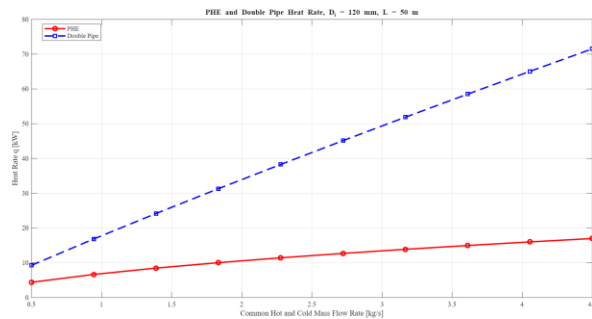


Figure9: PHE, Double Pipe Heat Rate Comparison

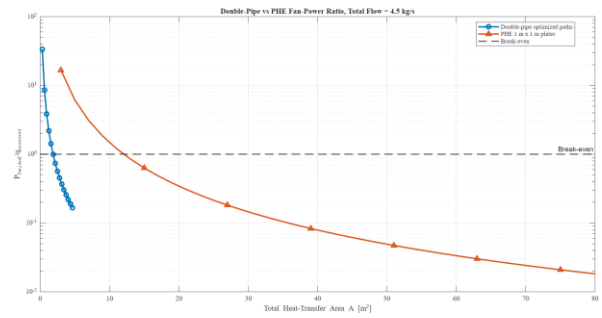


Figure 13: PHE, Double Pipe Fan-Power Ratio Comparison

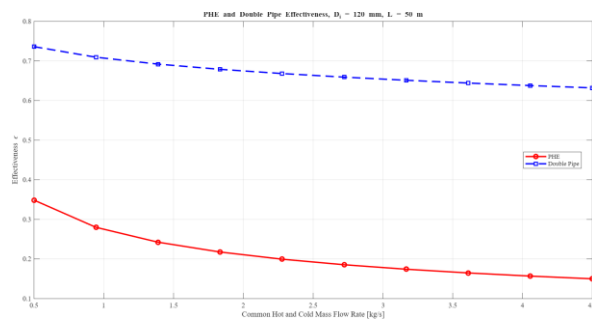


Figure 10: PHE, Double Pipe Effectiveness Comparison

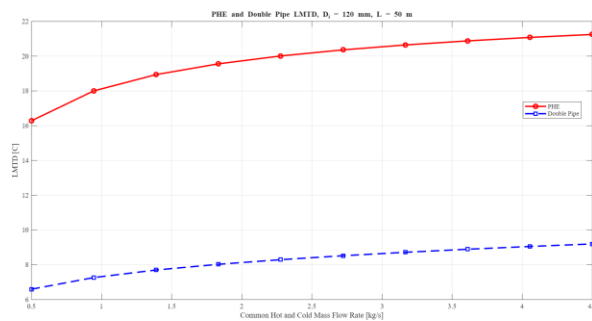


Figure 11: PHE, Double Pipe LMTD Comparison

**Sample Calculations (for plate in counterflow):**

Using the chosen  $\dot{m}_h = 1 \frac{kg}{s}$ :

For the hot fluid

$$D_e = \frac{2b}{\phi} \approx 2b = 0.02 \text{ m}$$

$$G_{c,h} = \frac{\dot{m}_h}{b * W_p} = \frac{1 \frac{kg}{s}}{0.01 \text{ m} * 0.3 \text{ m}} = 333 \frac{kg}{sm^2}$$

$$Re_h = \frac{G_{c,h} D_e}{\mu} = \frac{333 \frac{kg}{sm^2} * 0.03 \text{ m}}{1.81 * 10^{-5} \frac{Ns}{m^2}} = 551,933.$$

The Nusselt number correlations could then be defined as

$$Nu_h = \frac{h_h D_e}{k} = C_h (Re)^n Pr^{\frac{1}{3}} \left( \frac{\mu}{\mu_w} \right)^{0.17}$$

$$= 0.4(551,933)^{0.5} (0.715)^{\frac{1}{3}} * 0.9 = 265.$$

Therefore

$$h_h = 350 \frac{W}{m^2 K}.$$

The same process could be done with  $\dot{m}_c = 4.5 \frac{kg}{s}$  to produce

$$h_c = 744 \frac{W}{m^2 K}.$$

These values are likely high due to the small b value. The overall heat transfer coefficient can then be defined as

$$U = \frac{1}{\frac{1}{350 \frac{W}{m^2 K}} + \frac{0.01 \text{ m}}{15 \frac{W}{m * K}} + \frac{1}{744 \frac{W}{m^2 K}}}.$$

$$U = 205 \frac{W}{m^2 * K}$$

Then, further analysis can be done to determine the effectiveness, heat rate, and cold outlet temperature.

$$C_r = \frac{C_{min}}{C_{max}} = \frac{\dot{m}_h c_p}{\dot{m}_c c_p} = 0.222$$

$$NTU = \frac{U * A}{C_{min}} = \frac{205 \frac{W}{m^2 * K} * 10 \text{ m}^2}{1609.6 \frac{W}{K}}$$

$$= 1.27$$

Using the NTU and  $C_r$  the effectiveness can be calculated.

$$\epsilon = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}} = \mathbf{0.74}$$

Then the heat rate can be calculated using

$$q = \epsilon * C_{min} (T_{h,i} - T_{c,i}) = \mathbf{18018 \text{ W}}$$

Now knowing the heat rate, the energy balance can be used to calculate the outlet temperatures.

$$T_{h,o} = T_{h,i} - \frac{q}{\dot{m}_h * c_p} = 25^\circ\text{C} - \frac{18018 \text{ W}}{1609.6 \frac{W}{K}}$$

$$= 8.82^\circ\text{C}$$

$$T_{c,o} = T_{c,i} + \frac{q}{\dot{m}_c * c_p} = 0^\circ\text{C} + \frac{18018 \text{ W}}{4527 \frac{W}{K}}$$

$$= \mathbf{4.1^\circ\text{C}}$$